CNN Processing in Radiology: Data Preparation and Supervision in Knee MRI

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Abstract. Convolutional neural networks (CNNs) turn radiology images into learned features and predictions, but their usefulness depends on the images and labels used for training. We explain this process through a knee MRI case study of 4,407 examinations and 24,371 series. Only 58 examinations have expert labels for all twelve findings; the other 4,349 have reports without these labels. The audit documents multilingual reports, text corruption, missing series, and report–expert disagreements. These findings motivate slice-level feature extraction, fusion across imaging planes, explicit input masks, and learning from uncertain targets. We relate the workflow to radiology research and industry-supported imaging frameworks while separating measured data findings from proposed CNN methods. No trained model performance or clinical benefit is claimed.

Keywords: Radiology · Knee MRI · Convolutional neural networks · Dataset audit · Weak supervision.

1 Radiology Research and the Knee MRI Case Study

Radiology image analysis includes classification, localization, and segmentation. Classification assigns a finding to an image or examination; localization identifies a region, while segmentation outlines structures at the pixel or voxel level [6]. This paper concerns examination-level classification. We use the RSNA knee MRI dataset to study how image preparation and label quality shape that task [5].

An MRI examination contains several series and slices showing different parts of the anatomy. A CNN must connect those images to a finding assigned to the examination. The clinical report offers another description of the case, but it is not necessarily a complete label set. Our knee study makes these issues concrete; its measured prevalences and coverage do not describe radiology datasets in general.

MRNet offers a useful knee MRI precedent: it encodes slices with an ImageNet-pretrained AlexNet, pools features across slices, and combines series predictions. Its development labels came from reports, while its validation labels came from independent image review [1]. CheXpert also separates report-derived supervision from expert evaluation, although its chest-radiograph labels do not transfer directly to knee MRI [2]. Snorkel provides a framework for

learning from noisy labeling sources [3]. These works motivate checking label quality; their results are not results for the present dataset.

1.1 Industry research and clinical context

In NVIDIA's announcement of its Massachusetts General Hospital collaboration, Jensen (Jen-Hsun) Huang described deep learning as a way to extend physicians' capabilities [7]. That statement provides industry context, not clinical validation. NVIDIA researchers also co-developed MONAI with academic and clinical collaborators. Its published framework extends PyTorch with medical-image transforms, networks, and training and inference utilities [8]. This is relevant to our separation of preprocessing, CNN prediction, and evaluation; MONAI itself was not used in the recorded audit.

Google Health and DeepMind researchers, with clinical collaborators, evaluated a mammography system on UK and US data and in a radiologist reader study [9]. The work illustrates the distinction between model scoring and comparison with clinical readers. Its breast screening results cannot establish performance for our knee MRI task.

1.2 Case-study endpoint and image-only inference

The source competition scores twelve predictions per examination using mean finding-level ROC-AUC [4]:

$$\text{MacroAUC} = \frac{1}{12} \sum_{j=1}^{12} \text{AUC}_j. \tag{1}$$

Accuracy alone is misleading here. In the 58 expert examinations, positive rates range from 0.155 to 0.603. Always predicting negative for the MCL label alone gives 84.5 percent accuracy on that subset. This is a majority-class baseline, not a trained-model score.

Reports can supply training targets but are absent at test time. The source task requires offline inference with preloaded weights and dependencies, writing twelve scores per `StudyInstanceUID` to `submission.csv` [4]. These are case-study requirements, not universal radiology rules. Training and test prevalence may differ.

2 Audit Methods and Dataset Findings

We use recorded inspection results and the accompanying preparation code [5]. Examination identifiers connect reports, expert labels, and series metadata. The report review covers the 58 labeled examinations. Language detection uses `langdetect` on the first 400 characters; these are detector assignments, not manually verified languages. A full-partition pass reads one DICOM header per examination. A separate sample of 200 examinations and 1,088 series supports the field-strength and institution-field observations.

Table 1. Positive rate of each finding among the 58 expert-labeled examinations. These counts describe the released expert-labeled training subset.

Finding	Positive	Negative	Rate
Effusion	35	23	0.603
Synovitis	27	31	0.466
Medial meniscus	26	32	0.448
Anterior cruciate ligament	24	34	0.414
Lateral meniscus	23	35	0.397
Patellofemoral OA	21	37	0.362
Contusion	19	39	0.328
Fracture	18	40	0.310
Medial OA	15	43	0.259
Baker's cyst	12	46	0.207
Lateral OA	11	47	0.190
Medial collateral ligament	9	49	0.155

2.1 Images and labels

The corpus contains 4,407 examinations and 24,371 series, about 5.5 series per examination. Series in one sampled examination had 16–22 slices; this is not a corpus-wide range. The header pass took 76 seconds and found zero unreadable examinations. It did not validate every image slice. Vendor counts were Siemens 1,951, Philips 1,301, GE 905, Canon/Toshiba 226, and Fujifilm/Hitachi 24. In the metadata sample, 1.5 T and 3.0 T values occurred in roughly a 60:40 ratio.

Label coverage is all-or-nothing: 58 examinations have all twelve expert labels (Table 1); 4,349 have none. Missing means unknown, not negative. OA denotes osteoarthritis.

2.2 Languages and text quality

Detected languages are English 39.4 percent, Spanish 15.5, Turkish 12.4, Croatian 9.2, Greek 7.3, German 5.9, Bulgarian 5.0, Dutch 3.5, and French 1.8. The expert subset covers eight languages. French has no labeled examples, while German, Dutch, Greek, and Bulgarian each have only two to four. These small groups cannot support reliable language comparisons.

Section headers are inconsistent: *findings* appears in 19.5 percent of reports, *impression* in 15.3 percent, and its Spanish equivalent in 15.5 percent. These are separate frequencies, not a measured union. Processing must therefore accept whole reports rather than require a particular section. A date placeholder occurs 710 times in 594 reports.

A more consequential artifact affects 87 reports, with 228 occurrences of `intact` directly followed by a digit. The audit identifies a substitution of `0.` with `intact`, turning `0.9x0.4cm`, for example, into `intact9xintact4cm`. Standalone `intact` also occurs 2,975 times as a genuine negation cue. The normalization

Table 2. Examples of report–expert disagreement from the recorded audit. Paraphrased report statements are not equivalent to the provided labels.

Finding	Report states	Expert label
Baker's cyst	cyst reported absent	positive
Lateral meniscus	menisci described as normal	positive
Lateral OA	cartilage described as normal	positive
Medial OA	cartilage described as preserved	positive

repairs only digit-adjoining occurrences, preserving standalone uses. This text repair is not a label extractor.

Repairing a decimal restores readable text, not a diagnosis. Language detection and text normalization should remain separate from interpreting a finding and deciding whether it is a suitable training target.

2.3 Where reports and expert labels disagree

Table 2 shows recorded cases in which a report negates an expert-positive finding. Severity alone does not explain the reviewed cases either: mild effusion is positive in one examination, while moderate effusion is negative in another. These observations show why faithful report extraction cannot be assumed to recover expert labels exactly. We therefore do not build a rule-based label extractor. Without independent image adjudication, we cannot separate reporting omissions, differing criteria, translation ambiguity, and annotation error. Nor do these examples prove a numerical ceiling on text-model AUC: AUC measures ranking, not exact label recovery.

A correct reading of a negative report can still disagree with the expert label. Better negation handling cannot resolve that mismatch. Keeping report interpretation separate from expert comparison helps distinguish parsing errors from differences between the label sources.

3 Preparing Images and Separating Folds

The selector uses plane, fluid sensitivity, and fat suppression to retain one series in each of four slots (Table 3). Fluid-sensitive slots prefer fat suppression. Ties use slice count when supplied, then series identifier; the released series CSV has no slice-count field. The slot called sagittal T1 is selected as sagittal and non-fluid-sensitive, so its name does not independently verify the pulse sequence.

The loader clips intensities by percentile, rescales finite values, samples rounded evenly spaced slice indices, and resizes each slice with bilinear interpolation. Short volumes may repeat slices. Missing or unreadable slots become zero-filled arrays with separate presence masks. Fixed array depth does not standardize physical slice spacing, and matching indices across planes do not imply

Table 3. Canonical-slot coverage across the 4,407 training examinations. Coverage denotes availability under the metadata selection rule.

Canonical slot	Coverage (%)
Axial fluid-sensitive	100.0
Sagittal T1	96.8
Coronal fluid-sensitive	96.4
Sagittal fluid-sensitive	94.2

anatomical registration. Cached volumes are keyed by identifiers, preprocessing version, and settings; changed DICOM contents under unchanged identifiers require explicit cache invalidation.

Coverage in Table 3 describes metadata selection, not image decoding. Retaining the selected series identifier and loading outcome distinguishes missing acquisitions from read failures and links a model's prediction to the images actually supplied.

Institution identity is absent from the tabular release; station and institution names were also absent from all 1,088 sampled series. The grouping proxy uses manufacturer, scanner model, field strength, report language, and placeholder signature. Small groups are coarsened to manufacturer/language, then language, with remaining small groups merged deterministically. The recorded result has 24 groups, ranging from 13 to 590 examinations, with no singletons. Fold sizes are 869, 865, 870, 880, and 865 training-eligible examinations. All 58 expert examinations are excluded. For group g_i, fold f_i, expert set $\mathcal{E}$, and fold sets $\mathcal{F}_r$, the verified assignment satisfies

$$g_i = g_{i'} \implies f_i = f_{i'}, \qquad \mathcal{E} \cap \left(\bigcup_{r=1}^{5} \mathcal{F}_r \right) = \emptyset, \tag{2}$$

No proxy group crosses folds. This does not prove patient or hospital independence: the proxy can combine institutions or divide one institution. An examination must also remain intact; splitting its slices between training and validation would expose related images on both sides.

4 How CNN Processing Works in Radiology

A convolution applies the same learned filter across local image regions, producing a feature map. Nonlinear activations and successive layers combine these responses; spatial pooling summarizes local features. The learned representation feeds a task-specific prediction head [6]. In our case study, that head predicts findings for the whole examination, not a lesion outline or a complete radiology report.

During training, backpropagation uses the loss gradient to update network weights. At inference, the fitted network produces scores without updating those

weights. Pretrained encoders may be used as fixed feature extractors or fine-tuned on medical images [6]. Neither strategy removes the need to validate the model on the intended task. The present NumPy array-and-mask interface can feed a PyTorch CNN, but no trained checkpoint has been established [5].

4.1 From slices to findings

Let $x_{v,s}$ be slice s in slot v, with presence indicator m_v. A slice CNN $f_{\theta,v}$, aggregation operator A_s, and fusion head $h_{\phi,k}$ can produce finding score p_k:

$$z_v = A_s\{f_{\theta,v}(x_{v,s})\}, \qquad p_k = \sigma(h_{\phi,k}(\{m_v z_v, m_v\}_v)), \tag{3}$$

Here σ is the sigmoid. Separate outputs allow findings to co-occur. The audit does not choose shared or separate encoders. Repeating slices can alter mean pooling; duplicating identical deterministic features leaves maximum pooling unchanged. Neither recovers evidence missed by sampling. An examination-level positive label also does not make every slice positive.

Spatial pooling within a slice and aggregation across slices serve different purposes. The former reduces image features; the latter connects them to the examination label. Plane fusion then combines the available views. It must not assume that matching array positions show matching anatomy. These choices require comparison on expert-labeled examinations.

The fusion head must use the presence mask: a CNN can produce nonzero features from zero inputs because of learned biases. Masking unavailable features and passing availability to the head makes missing acquisitions explicit. Examinations with no readable slot need a failure path, not a normal prediction. Scanner appearance may remain in images even when scanner metadata is excluded, so grouped folds alone do not prove that a network learns pathology rather than acquisition cues.

4.2 Soft targets and trust weights

For a soft report target $q_k \in [0,1]$, CNN logit a_k, and $p_k = \sigma(a_k)$, binary cross-entropy gives

$$\ell_k = -q_k \log p_k - (1-q_k)\log(1-p_k), \qquad \frac{\partial \ell_k}{\partial a_k} = p_k - q_k. \tag{4}$$

Wrong targets change the CNN gradient even when images load correctly. Unknown targets must be masked. Soft probabilities express uncertainty without guaranteeing calibration to expert labels.

For examination i and finding j, let q_{ij} be the report target, p_{ij} the CNN output, and $\widehat{A}_j$ report–expert AUC. The trust rule combines target certainty with agreement:

$$w_{ij} = \min\Big(1, \max\Big(\epsilon_w, 2|q_{ij} - 0.5| \max(0, 2\widehat{A}_j - 1)\Big)\Big). \tag{5}$$

Here $0 < \epsilon_w \leq 1$ sets a minimum weight. AUC at or below chance receives that floor. Distance from 0.5 is a heuristic, not a calibrated probability of correctness, and the two signals need not be independent. This rule requires agreement estimates; none are reported as measured results in this audit.

With observation mask o_{ij} and positive-class weight α_j, the training objective is

$$\mathcal{L} = -\frac{\sum_{i,j} o_{ij} w_{ij} \left[\alpha_j q_{ij} \log p_{ij} + (1 - q_{ij}) \log(1 - p_{ij})\right]}{\sum_{i,j} o_{ij} w_{ij}}. \tag{6}$$

This applies when the denominator is positive. Unknown targets are omitted from both sums; a batch without observed targets gives no supervised update. Probabilities are bounded away from the endpoints for numerical evaluation. Class weights address frequency imbalance; trust weights address label reliability. Neither their settings nor their benefit has been measured here.

Increasing α_j raises the positive term; changing w_{ij} changes the target's contribution as a whole. A rare finding is not necessarily unreliable. Separate experiments are needed to distinguish the effects of frequency correction and label-source weighting.

5 Statistical Evaluation and Limitations

The agreement evaluator compares finite report scores q_{ij} with observed expert labels $y_{ij} \in \{0, 1\}$ on a usable set $\mathcal{I}_j$. Its positive and negative subsets have sizes n_j^+ and n_j^-. With ascending score ranks R_{ij} and average ranks for ties, empirical AUC is

$$\widehat{A}_j = \frac{\sum_{i \in \mathcal{P}_j} R_{ij} - n_j^+ (n_j^+ + 1)/2}{n_j^+ n_j^-}. \tag{7}$$

Here $\mathcal{P}_j$ is the positive subset. AUC is undefined if either class is absent. A mean over only defined findings must name that subset; it is not necessarily the twelve-label score in Equation 1.

The signed mean-score offset on the same examinations is

$$\Delta_j = \frac{1}{|\mathcal{I}_j|} \sum_{i \in \mathcal{I}_j} (q_{ij} - y_{ij}). \tag{8}$$

It compares mean predicted probability with the expert-positive fraction, not thresholded positive rates. It is only one aspect of calibration. Usable counts and exclusions must accompany any estimate.

AUC measures ranking; the offset compares averages; a threshold turns scores into decisions. Good ranking does not establish calibrated probabilities, and a small offset can hide errors that cancel across examinations. None of these summaries alone establishes clinical utility.

The bootstrap samples examinations with replacement. Let $\widehat{A}_j^{*(b)}$ be resample AUC, $\mathcal{B}_j$ contain resamples with both classes, and Q_u denote an empirical quantile. The percentile interval is

$$\mathrm{CI}_{j,0.95} = \left[Q_{0.025}\left(\{\widehat{A}_j^{*(b)}\}_{b \in \mathcal{B}_j} \right), Q_{0.975}\left(\{\widehat{A}_j^{*(b)}\}_{b \in \mathcal{B}_j} \right) \right]. \tag{9}$$

The configured default is 1,000 seeded resamples, not a completed analysis. Single-class resamples are omitted; if none remain, the interval is undefined. Such intervals describe sampling variability, not annotation accuracy. No measured AUC, offset, or confidence interval is available.

Only 9–35 positives per finding are available in the expert subset. Using these 58 examinations to select prompts, trust weights, thresholds, or calibration makes them development data; later scores on the same examinations are not independent validation. Image-only, text-only, and combined systems also use different information. Comparisons must state the inputs, split, and reference-label source. Any accuracy result must identify its split and label set. A report-assisted score cannot stand in for image-only test performance.

The audit does not establish CNN discrimination, clinical utility, or the best architecture. The metadata runtime is one recorded pass, language and acquisition groups are inferred, and the report examples have not been independently adjudicated. No external or prospective validation is reported. These limits leave a clear next step: evaluate image models and supervision choices on independent expert annotations.

5.1 From a model score to a radiology study

A useful model comparison fixes the input selection, examination split, and reference labels while changing pooling, masking, or supervision. Each run should record its encoder, pretrained weights, resolution, slice count, optimizer, schedule, stopping rule, and hardware. A later reader study would ask a different question: whether showing predictions changes radiologists' decisions. The intended use and reference standard must be stated for each study. Neither an industry platform nor a strong benchmark score answers that clinical question by itself.

6 Conclusion

CNN processing in radiology connects image preparation, learned features, aggregation, and task-specific predictions. The knee MRI audit shows how missing series and uncertain labels affect that chain. Industry frameworks and published imaging studies provide context, while the measured findings support explicit masks and careful supervision. Model performance and benefit to radiologists remain questions for independent evaluation.

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